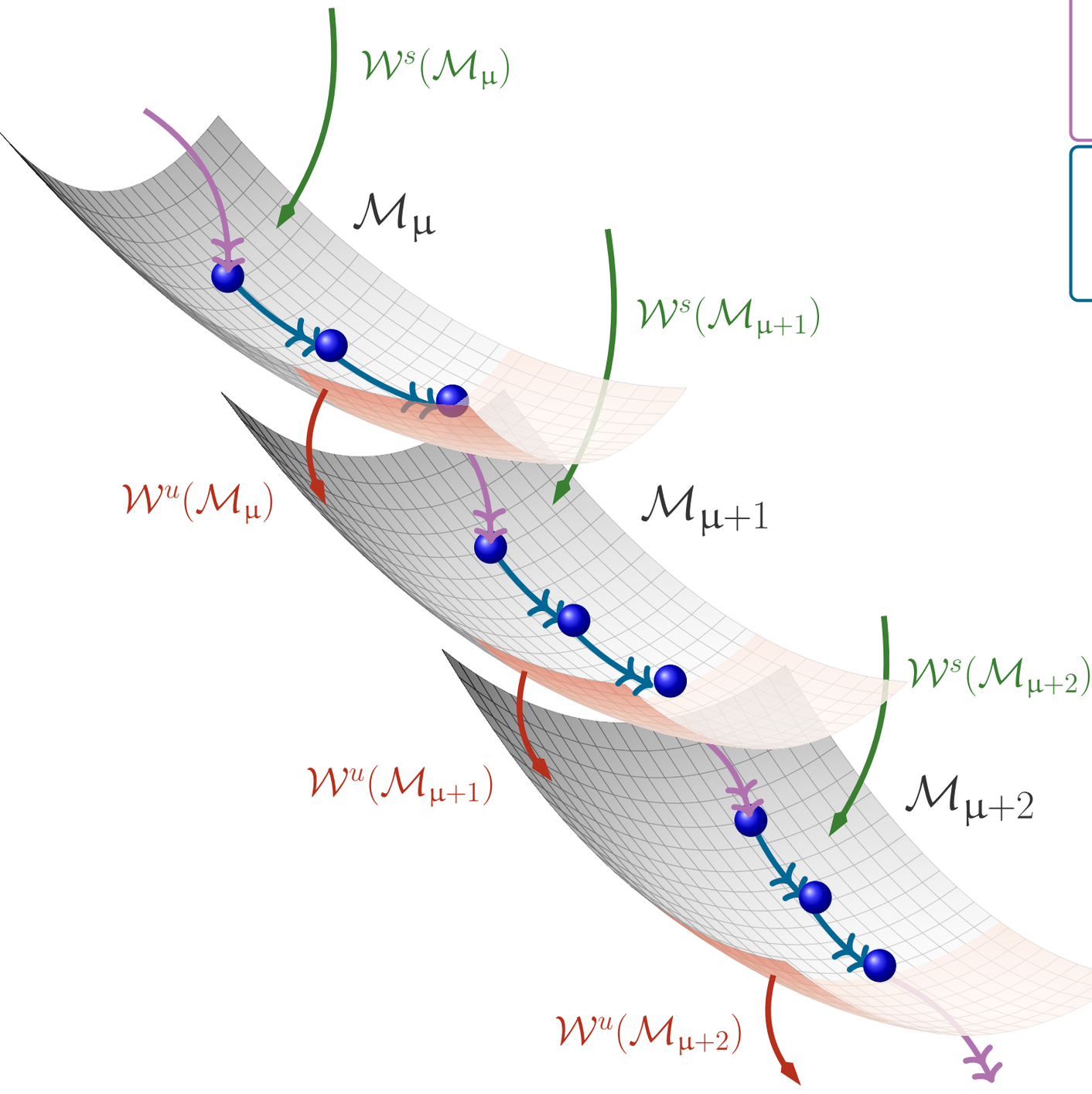
Memory transitions in the Hopfield model



Fast (Hopfield) dynamics:

$$\tau_x \dot{x} = -x + \mathbf{W}(z)\Phi(x)$$

Slow (saliency) dynamics:

$$\tau_z \dot{z} = -z + \kappa \mathbf{T}\Phi(x)$$

$\mathcal{M}$: **slow (memory) manifolds**

$\mathcal{W}^s(\mathcal{M})$: **attracting manifolds**

$\mathcal{W}^s(\mathcal{M})$: **repelling manifolds**